ST MARY'S DSG, KLOOF

PHYSICAL SCIENCE CLASS TEST
KINEMATICS

GRADE: 10
TIME: 40 MINS
ASSESSOR: T. Kirsten

14 August 2023
TOTAL: 43
MODERATORS: K. Bolton and G. Main

NAME: _____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 6 pages.
2. Answer on the question paper in the spaces.
3. Write your name on your question paper.

You may need the following equations:

$$v = \frac{\Delta x}{\Delta t} \qquad v_f = v_i + a\Delta t$$

QUESTION 1 MULTIPLE CHOICE – Place a X over your choice of letter on the grid below:

1.1	A	X B	C	D
1.2	A	B	X C	D
1.3	A	B	C	X D

1.1 Which of the following combinations contain both scalar and vector quantities?

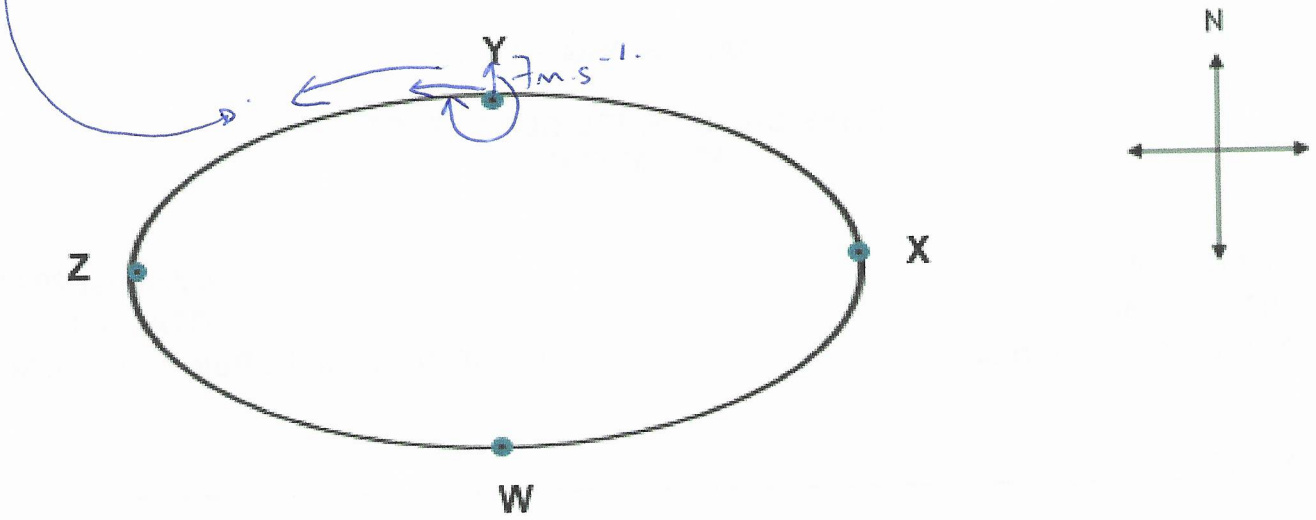
A mass, temperature, time

B mass, speed, force

C displacement, velocity, momentum

D distance, speed, time

1.2 Refer to the following diagram. An Olympic 400 m athlete starts from rest at point W, runs **anti-clockwise**, and accelerates until she reaches point X. She then runs at a **constant velocity** of 7 $m.s^{-1}$, past point Y, until she reaches point Z. At point Z she accelerates again until she finishes at W.



The instantaneous velocity of the athlete at point Y is:

A 7 $m.s^{-1}$ North

B 7 $m.s^{-1}$ South

C 7 $m.s^{-1}$ West

D 7 $m.s^{-1}$ East

1.3 If a car accelerates at 4 $m.s^{-2}$, it means that:

A Every 4 seconds the car travels 1 m further.

B Every second the car travels 4 m more.

C Every second the car travels 4 m faster.

D Every second the car travels 4 $m.s^{-1}$ faster.

3 x 2 = [6]

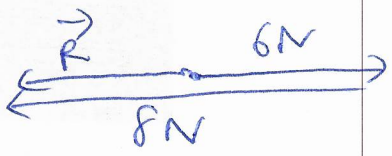
QUESTION 2

2.1 Define *resultant vector*. (2)

Single vector that has the same effect as original vectors combined

2.2 For the following situations, draw a sketch and **calculate the resultant vector**, using the tail-to-head method of vector addition:

2.2.1 Two boys simultaneously push on a box. Jack pushes to the right, with a force of 6 N. John pushes to the left with a force of 8 N. (3)

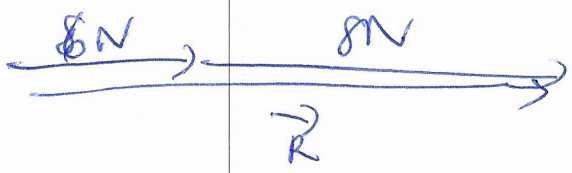


$\vec{R} = 6 + (-8)$

$= -2N$

$= 2N$ left

2.2.2 John then comes to the same side of the box as Jack, and they both push on the box to the right, Jack still pushing with a force of 6 N, and John 8 N. (3)



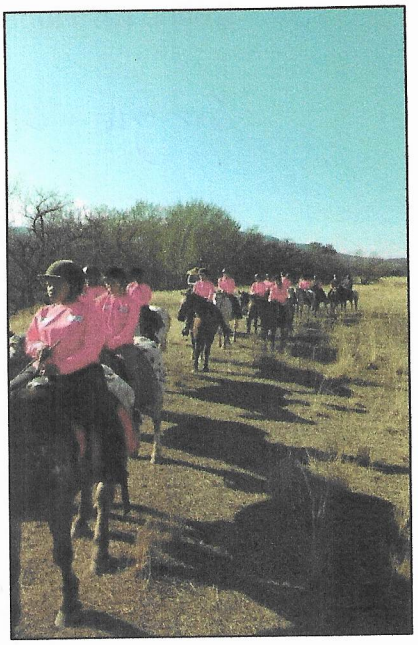
$\vec{R} = 14N$ right.

[8]

QUESTION 3

Some Grade 10 girls set out on a gentle horse ride on The Journey.

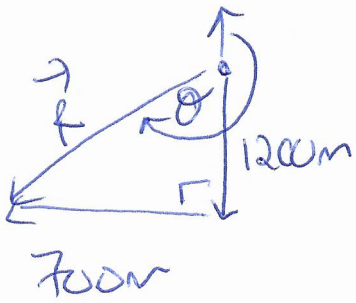
- The horses and riders initially walk for 1,2 km South and stop to admire the view.
- They then turn West, and the horses trot for 700m, then stop at a waterfall.



3.1 Define *displacement*. (2)

Change in position

3.2 Sketch (i.e not to scale) a labelled vector diagram showing the route of the horse ride, including the resultant displacement from the starting point. (3)



3.3 Calculate the resultant displacement of horses and riders from the starting point. (5)

$\vec{R} = \sqrt{1200^2 + 700^2}$

$= 1389,24$ m

$\tan\theta = \frac{700}{1200}$

$\theta = 30,26°$

∴ on a bearing $30,26 + 180 = 210,26°$

3.4 From their position 700 m West, the horses and riders then turn and trot directly back to the starting point, following the shortest route in a straight line.

3.4.1 Define *distance*. (2)

actual path length travelled.

3.4.2 If the entire ride took 30 minutes, calculate the average speed of the horses during the ride. (4)

1200 m + 700 m + 1359,24 = 3259,24
shortest route

$$\text{Speed} = \frac{3259,24}{30 \times 60} = 1,83\ m\cdot s^{-1}$$

3.5 Define *velocity*. (2)

Rate of change of displacement.

3.6 Determine the average velocity of the horses from the starting point of the ride until the horses returned. (2)

$$V = \frac{\Delta x}{t} = \frac{0}{1800} = 0$$ → back @ start ∴ $\Delta x = 0$.

3.7.1 Define *acceleration*. (2)

Rate of change of velocity.

3.7.2 When the horses start to trot West, they accelerate from rest and reach a velocity of 1,8 $m.s^{-1}$ after 6 seconds. Calculate the acceleration of the horses. (4)

$V_i = 0$ ⊕ west.

$V_f = 1,8$

$t = 6$

$V_f = V_i + at$

$1,8 = 0 + a(6)$

$a = 0,3\ m.s^{-2}$ WEST.

3.8 The horses continue at a **constant speed** of 1,8 $m.s^{-1}$ for 5 minutes. Calculate the distance travelled in this time. (3)

$a = 0$

$V = 1,8$.

$t = 5 \times 60$

$= 300s$.

$\Delta x = v_i t + \frac{1}{2}at^2$

$= (1,8)(300) + 0$.

$= 540m$

[29]